

Flighty Travel Profile: Mobile Redesign PRD

01 Executive summary

Recommendation: extend Passport into an actionable cross-airline travel record, with editable Trips as its organizing unit. Help travelers retrieve a complete journey, inspect the flights behind a statistic, and reuse relevant history when preparing for another booked flight. Preserve operational tracking as the primary travel-day experience. Treat loyalty context as a separate validation pilot, not the reason the core release must succeed.

The initial Trip Memory idea addresses a recognizable organization need. Public evidence does not establish that photographs or journaling alone improve retention or conversion. Nor does it support immediately turning Flighty into a rewards hub: Triplt and AwardWallet already aggregate travel or reward information, and point.me addresses redemption decisions. Cross-airline coverage is useful positioning, but is not an exclusive advantage. ^{[6][7][8]}

The proposed **Flighty Travel Profile** is an internal concept name. Use familiar labels in the app: Passport, Trips, Your travel, and Before you fly. The core consists of conservative grouping, explicit corrections, linked statistics, and a relevant previous-trip reference. One note and an optional cover photo preserve the emotional layer without requiring content creation.

Decision	Proposal
Primary user	An iPhone traveler using multiple airlines, with recurring journeys or multi-leg international travel and some existing flight history.
Core job	Help me understand and retrieve my travel across airlines, and carry useful context into my next trip.
MVP	Editable Trips; bounded grouping; statistic-to-flight drill-down; related-history reference; notes; provenance; export and recovery.
Separate pilot	Manual loyalty program/status snapshot and a personal review reminder. No automatic balance or qualification calculations.
Defer	Connected reward accounts, credit-card data, award search, status forecasts and pre-booking recommendations.
Evidence level	Desk research supports organization and mixed paid-value concerns. Incremental retention, payment and loyalty demand remain hypotheses.
Document status	PRD v1.0, proposed 10 September 2026. Ready for design and engineering estimation; launch depends on the specified feasibility and validation gates.

02 Background, scope and decision log

This is an independent **mobile app redesign case study**, not a Flighty announcement. Scope: an iPhone extension, initially in English. Existing flight tracking, Friends, search, operational notifications and Passport form the baseline. No access to Flighty code, contracts, internal metrics, roadmap or customer database is assumed.

The earlier concept grouped flight segments into memories. This revision tests whether a persistent record creates practical value across travel cycles. It does not assume that a loyalty dashboard is the correct answer. A profile that only repeats current totals would also fail the differentiation test.

ID	Assumption / decision	Consequence
A01	Stable account-scoped flight IDs are available internally.	Engineering must confirm identity, field coverage and sync before implementation.
A02	Existing users are the first validation cohort.	Acquisition is secondary; recruit against current workflows.
A03	Initial research cohort: 6+ flown segments and 2+ operating airlines in the last 12 months.	A recruitment rule, not a factual definition of frequent travel or a runtime restriction.
A04	Trips and profile drill-down can reuse one history layer.	Reference source flights; do not duplicate the flight database.
A05	English artifact and UI copy follow the previous case-study deliverable.	Localization keys support future adaptation.
D01	Choose direction C: actionable cross-airline profile.	Trips support retrieval and reuse; memory is optional.
D02	Loyalty is behind an independent pilot flag.	Weak loyalty adoption cannot block core Trips.
D03	All success thresholds are proposals.	No claimed market-share, conversion or churn improvement before measurement.

03 Research method and evidence boundaries

Research checked current public product/help pages, first-party app descriptions, four Flighty Reddit discussions, one historical Flighty App Store review and two AwardWallet Google Play reviews. Airline qualification rules were examined to test feasibility. Sources were accessed on 10 September 2026. Public pages may describe capabilities absent from older screenshots or particular app versions.

The qualitative ledger contains **12 selected records**: nine Reddit accounts across four threads, one Flighty App Store reviewer and two AwardWallet Play reviewers. Repeated copies of the same review count once. Promotional replies and related-post recommendations are excluded. This is purposive desk research, not a complete scrape, survey, sentiment benchmark or representative prevalence estimate.

Reddit relative ages are preserved rather than converted into invented dates. The Passport thread has inconsistent crawl/relative-age metadata, so its exact publication date is unresolved. The App Store grouping review is dated January 3, 2024 and also includes an outdated Watch request. It establishes a historical need, not absence of every feature in the latest build. No interviews or usability sessions were conducted for this revision.

Label	Meaning
Public capability	A publisher documents a feature; this does not establish reliability, usage or universal coverage.
Observation	A specific user reports an experience; this is not a population frequency.
Judgment	An interpretation used for prioritization; it can change with better evidence.

Label	Meaning
Hypothesis	An expected behavioral or commercial effect that must be tested.
Requirement	Chosen behavior for this redesign, not a description of Flighty today.

No credible applicable market-share denominator or internal Flighty retention baseline was established. This report therefore does not invent TAM, revenue or share-growth estimates. Measure eligible active accounts internally, then exposure, activation and repeat use.

04 Existing Flighty baseline

Current capability	Evidence	Redesign implication
Operational tracking	Flighty documents inbound-aircraft tracking and delay prediction. ^[1]	Keep it primary. Airlines always being earlier or more accurate is not established.
Connection Assistant	Estimates and guidance cover international and multi-carrier connections. ^[2]	Do not position a second connection-risk engine as new. Reuse the existing assistant.
Passport	Route map, distance, time, aircraft and delay history are documented. ^[3]	New value is editable boundaries, retrieval and reuse, not basic totals.
Connection/layover counting	Existing rules distinguish short connections, layovers and roundtrips. ^[4]	User-owned Trip grouping must not silently rewrite existing visit statistics.
Free/Pro	Public help lists free Passport/tracking and paid advanced features; import/history access varies. ^[5]	Reuse actual entitlement checks; do not relabel paid imports as free.
Visual reference	The earlier case-study artifact records globe + sheet, flight rows, year selection and list/map sorting.	Treat as a historical design reference, not a current-build audit.

Before implementation, inspect the current iPhone build for trip editing, compound filters and linked statistics. Official documentation does not prove absence of undocumented features. If these have since shipped, implement only the remaining gap and revise the case-study claim.

05 Competitive landscape

The landscape is organized around the job each product owns. Strengths summarize public positioning; boundaries are judgments relative to this proposal. Cross-airline does not mean every airline, field or booking is supported.

Product / audience	Owned job and strength	Boundary / implication
United / United travelers	Booking, boarding, seats/bags, connections, MileagePlus and live updates. ^[9]	Carrier execution is authoritative. Complement it with independently booked cross-airline history.
Fly Delta / Delta travelers	Booking, boarding pass, upgrades, bags and disruption rebooking. ^[10]	Deep carrier workflow. Reviewed Android listing does not verify iOS parity.
American / AA travelers	Booking, check-in, rebooking, Live Activities and loyalty progress. ^[11]	A strong free alternative; history must save additional work.

Product / audience	Owned job and strength	Boundary / implication
ANA / Japan-linked travelers	Booking/status and Mileage Club earning, redemption and premium services. ^[12]	AMC web ecosystem reviewed; not a full mobile audit. Program-specific data matters.
FlightAware / travelers and aviation users	Live flight and airport tracking. ^[13]	Operational visibility is crowded; the reviewed page does not establish editable personal trip context.
Flightradar24 / enthusiasts	Aircraft exploration, flight history and map/filter subscription features. ^[14]	Aviation observation differs from personal journey retrieval. Avoid competing through telemetry volume.
Tript / organized and business travelers	Travel organization, alerts, alternatives, maps and Point Tracker. ^[6]	Already crosses itinerary and rewards categories; a trips list plus points tab is insufficient.
AwardWallet / points users	Balances, elite status, certificates, expirations and travel plans. ^[7]	Direct wallet incumbent. Connection upkeep is a recurring burden; do not rebuild aggregation in MVP.
point.me / redemption seekers	Reward-flight discovery and redemption guidance. ^[8]	Requires current availability. Route history is not an award recommendation.
Seats.aero / award-search users	Specialist award ecosystem; documentation explains dynamic versus saver pricing. ^[15]	Precise coverage and plan limits were not verified. No assumed integration.
Google Flights / fare shoppers	Multi-city search, date/price tradeoffs and booking handoff. ^[16]	Strong pre-booking comparison; not a personal loyalty ledger.
KAYAK / broad travel shoppers	Search, price alerts and account/Trips support. ^[17]	Public Trips page exposed little detail. Other coverage remains unverified.
Chase / eligible cardholders	Ultimate Rewards travel redemption and partner transfer options. ^[18]	Issuer owns balances and card-specific eligibility. Flight history cannot infer them.
Amex Travel / eligible cardmembers	Travel booking and Membership Rewards context. ^[19]	Issuer-specific offers and conditions; not a universal wallet.
Capital One Travel / adjacent candidate	Included in initial research scope.	Relevant pages could not be read. No capabilities or prices were scored.
App in the Air / historical	Flighty reports permanent closure on October 19, 2024. ^[20]	Historical migration reference, not an active competitor; source is a competitor. No inferred reason for closure.

Judgment: tracking, trip organization and rewards aggregation are occupied categories. The narrower opportunity is making already-maintained personal history editable, inspectable and reusable in a familiar mobile workflow. Defensibility is modest: continuity and low-effort reuse may help, but competitors can copy the interface. Export remains essential to trust.

06 User evidence ledger

ID / source	Observed evidence	Interpretation / limit
E01 · Aimee1980 · App Store · 2024-01-03	"I would love to be able to save individual trips." ^[21]	Historical grouping request from cabin crew; task specificity is strong, persona overlap partial.
E02 · MasterPh0 · Reddit · 3mo	"will be using as a flight log/diary." ^[22]	Carrier-app substitution can leave history as residual value; not universal.
E03 · Shmallyn · same thread · 3mo	Describes Flighty delivering gate/delay updates before airline apps. ^[22]	Counterevidence to abandoning operational differentiation.
E04 · Substantial_Nail8075 · Reddit · 13d	Requests tapping carrier statistics to inspect flights in a period. ^[23]	A specific retrieval problem; exact posting date unresolved.
E05 · Hyperspace-Hole · same thread · 12d	"Definitely agree that all stats should be navigable." ^[23]	Supports drill-down; not an independent study.
E06 · ZeroOpossum · Reddit · 8mo	"I'd love to use this app but the price is a barrier to entry." ^[24]	Interest is not willingness to buy a new feature.
E07 · scorch07 · same thread · 8mo	"I pay for a week or two when I have a trip and then revert back to free in-between." ^[24]	Episodic paying may be rational; off-trip opens alone do not justify annual payment.
E08 · Reviewer with employer Triplt Pro · same thread · 8mo	Employer provides Triplt Pro; reviewer questions renewing Flighty. ^[24]	Paid third-party tools can substitute too.
E09 · That-Camera-Guy · Reddit · 1y	"Old photos can be useful too" for reconstruction. ^[25]	Memory cues matter; this does not establish demand for a journal.
E10 · Imaginary_Check_9480 · same thread · 1y	Uses photos, journals and messages to reconstruct old flights. ^[25]	History can be incomplete; precise dates must not be invented.
E11 · John Chyan · AwardWallet Play · 2024-11-19	Reports account-update failures and price dissatisfaction. ^[26]	One historical integration complaint, not a service-wide error rate.
E12 · L D · AwardWallet Play · 2024-06-13	Values points, account and expiration consolidation. ^[26]	Counterbalances E11: a real wallet job exists, but in a specialist workflow.

Selected coding counts: organization/drill-down 3; mixed airline substitution/operational value 2; price or alternative-service value 3; historical reconstruction 2; wallet benefit/friction 2. These are counts of selected records, not complaint prevalence. No record directly validates paying Flighty for manual status forecasts. Earlier PRD participant totals are not repeated without recoding.

07 User segments, persona and jobs

Segment	Workflow / pain hypothesis	Repeat or paid value / priority
Casual traveler	Airline app + Wallet; infrequent travel, little history.	Travel-week payment can be enough. Secondary.
Frequent domestic, one airline	Carrier app + loyalty dashboard; familiar routes.	Reliability and retrieval may help; strong airline substitute. Control cohort.
Multi-airline frequent traveler	Several carrier apps + Flighty; fragmented history.	Retrieve and reuse context without reconstruction. Primary.
International frequent traveler	Multi-leg journeys and missing ground segments.	Editable boundaries and visible gaps. Primary scenario subset.
Points enthusiast	Airline accounts + AwardWallet + redemption search.	Accuracy matters; manual snapshots may be inferior. Pilot only.
Business traveler	Employer booking tool + airline execution + records.	Repeat routes and reviewable work trips; constrained booking choices. Primary subset.

Composite persona: Alex, a multi-airline frequent traveler. Alex takes approximately one return journey per month, mixes domestic work travel with international trips, and chooses schedule and cost before loyalty. This is a constructed design persona, not an interview participant. Alex remembers journeys by destination and purpose, not flight number, and wants the previous itinerary and a useful personal note when traveling again. Alex will not maintain multiple detailed reward ledgers just to improve a tracking app.

- Retrieval job: When related flights are scattered, help me find the whole journey and its original records so I can answer a travel question without rebuilding it.
- Preparation job: When a booked flight resembles prior travel, show the relevant trip and my note so I can reuse experience while checking current operational information separately.
- Trust job: When grouping is uncertain, explain it and let me correct it so the record stays mine.
- Emotional job: Let me attach a small personal cue after meaningful travel without requiring a scrapbook.

Willingness to use or pay is unmeasured in each segment. Recruitment must include former Flighty users and people satisfied with airline apps, not only enthusiasts.

08 Strategic alternatives and prioritization

Planning scores use 1-5, with 5 favorable. Weights: user value 25%, differentiation 15%, retention 20%, acquisition 10%, subscription 10%, feasibility 15%, brand fit 5%. These are explicit judgments, not measured market facts.

Direction	Value	Diff.	Ret.	Acq.	Sub.	Feas.	Fit	Weighted
A Better tracker	5	2	4	3	4	3	5	3.75
B Memory/journal	3	2	2	3	2	5	3	2.85
C Actionable profile	4	3	4	3	3	4	5	3.70
D Loyalty intelligence	4	3	4	3	4	2	3	3.40
E Rewards hub	3	1	4	2	3	1	2	2.45
F Pre-booking intelligence	4	3	3	4	4	1	2	3.10

Direction	Technical / data dependencies	Privacy, partnerships, defensibility
A	High for new prediction quality; operational feeds and model evaluation.	Travel data, feed contracts; existing operational expertise can defend value.
B	Low-medium: grouping, media and sync.	Private notes/photos; few partners; easy to copy.
C	Medium: identity, grouping, aggregation and lineage.	Private history; internal data access; continuity helps but no hard moat.
D	High: current rules, credited program, fare basis and balances.	Account context and licensed connections or manual work; weak without reliable coverage.
E	Very high: account connectivity and reconciliation.	Broad sensitive data, commercial access and entrenched competitors.
F	Very high: offers, eligibility, earning rules and preferences.	Spending consequences, multiple providers and high accuracy requirements.

Choose C for this redesign; retain A as the company's core investment. A slightly outscores C and both are effectively tied given subjective inputs. This case has no access to improve flight feeds. C is the most plausible adjacent UX experiment using current records. If stable identity is unavailable or research finds little retrieval pain, reduce scope to a narrower Passport drill-down improvement.

The cross-airline-home framing is supported only as a limited travel-record hypothesis. A broad frequent-traveler super app and loyalty-led paid proposition remain unvalidated.

09 Positioning, business logic and principles

Positioning: Flighty makes your flying understandable across airlines: know what is happening now, find the trip behind your history, and bring useful context to your next journey. The incremental proposition is history that is correctable and useful within an app already used for flights.

Business hypothesis	Mechanism	Required evidence
Retention	Reduced reconstruction effort creates reuse across journeys.	Second meaningful action and subsequent-trip return versus control.
Conversion	Useful history preview increases demand for existing Pro automation/depth.	Incremental paid conversion among free users; no forced history lock.

Business hypothesis	Mechanism	Required evidence
Acquisition	User-selected trip summaries explain the value to others.	Attributed new-user activation; sharing is V2.
Switching friction	A curated record becomes useful over time.	Durable usefulness with export available.
Frequency	Relevant pre/post-trip actions add occasional utility.	Useful actions outside travel days; daily opens are not the goal.

- Operational priority: active-flight information remains easier to reach than profile content.
- Evidence before advice: past flights do not prove current reliability or benefit eligibility.
- User authority: confirmed boundaries and notes survive regrouping.
- Progressive setup: deliver value before asking for home airports or loyalty.
- Source visibility: statistics have reproducible contributing-flight lists.
- Honest incompleteness: unknown values and missing history remain visible.
- Low maintenance: routine reward re-entry is not required for core value.
- Portable ownership: correction, export and deletion are part of the contract.

10 MVP, non-goals and hypotheses

ID / priority	Deliverable	Hypothesis / boundary
F01 P0	Passport entry and contextual introduction	Explains value without slowing flight access.
F02 P0	Trip suggestions and manual creation	Conservative suggestions reduce organization effort.
F03 P0	Merge, split, rename, remove and undo	Ambiguity is correctable without data loss.
F04 P0	Trip detail and statistic drill-down	Users retrieve known trips and explain totals.
F05 P0	Related-history preparation reference	History has utility beyond retrospective browsing.
F06 P0	Short note; optional single cover image	Light personalization aids recognition.
F07 P0	Provenance, privacy, export and sync recovery	Record ownership sustains trust.
F08 P0	Instrumentation and feature controls	Core effects can be measured independently.
L01 P1 pilot	Manual program/status snapshot and review reminder	Minimal loyalty context is worth maintaining; no inferred qualification.
V2	Alliance breakdown, sharing, richer notes, one licensed connection trial	Only after core utility and quality gates.
Future	Qualification scenarios and pre-booking decision support	Requires verified rules, fare/crediting data and available offers.

Non-goals: booking/payment, automatic check-in, live rebooking execution, a new risk engine, full hotel itineraries, expense management, social feed/badges, autonomous transfers, airline-password storage, banking connections, guaranteed upgrades/lounge access/status, and daily engagement nudges. Loyalty

and Wallet do not become permanent tabs.

11 Information architecture and mobile screens

Preserve the current production navigation. Add an “Overview / Trips” local switch inside Passport. Overview retains existing statistics and gains supported drill-down. Trips contains proposals and saved journeys. A subordinate Before you fly row on the upcoming-flight surface opens relevant history without covering operational alerts.

Screen	Purpose / hierarchy	Interaction and states
S01 Flights/Home	Active flight/disruptions first; upcoming flights next; one eligible history cue below.	Open flight. No upcoming flight keeps Search prominent. No match means no cue.
S02 Passport Overview	Year, existing totals, supported airline/route drill-down, data coverage.	Tap a statistic. Loading, empty, partial history, stale sync.
S03 Trips	At most 3 visible proposals, then saved Trips; Upcoming/Past and title search.	Review/create. Empty, pending, saved, no results, offline.
S04 Review Trip sheet	Proposed title/dates, flight rows, reason; Save Trip, Edit, Not a Trip.	Select/remove legs; explicit confirmation. Invalid selection/conflict.
S05 Trip Detail	Title/dates/status; route; next flight; full timeline; totals; note and optional image.	Open a leg or edit. Partial stats, gap, cancellation, completion.
S06 Matching Flights sheet	Filter chips, denominator and exact source rows with linked Trip labels.	Open flight/Trip. Each flight appears once.
S07 Before You Fly sheet	Reason, one prior same-direction route, personal note/date, current-flight link.	Open historical Trip or dismiss. No match means absent.
S08 Loyalty pilot sheet	Program/status, user-updated date, optional balance and review date.	Edit or open official site. Unverified/outdated/unavailable.

A map is supplementary: all routes and actions have text equivalents. Native Back restores filter and scroll position. Visible menu actions duplicate swipe actions. No chatbot is required.

12 End-to-end journeys

Journey / trigger	Sequence	Completion / alternative
Existing user opens Passport	Show one optional explanation; process available personal history; show a proposal with source flights; review; save.	Saved Trip is visible immediately. Skip returns to existing Passport; loyalty is not requested.
New user has little history	Add flights through existing permitted methods; show a sample clearly labeled Example; offer manual Trip creation after one personal flight.	No fabricated personal insight. Fewer than 6 flights does not block manual use.
International trip recognition	Review PIT-ORD-HND, onward Japan flight and return; inspect dates; confirm boundary; name the Trip.	Treat route examples as illustrative; KIX-PIT is a journey shorthand, not an asserted nonstop service. Every actual connection is a separate record.
Open-jaw correction	PIT-HND and KIX-SFO-PIT are proposed with a visible HND-to-KIX gap; user confirms one trip or splits it.	Gap is labeled Travel not recorded. Do not invent a train or flight.
Merge / split	Open Edit; select adjacent Trips or a split boundary; preview resulting titles, flights and notes; save atomically.	Undo is available; source flight records and lifetime totals do not change.
Next booked flight	A same-direction previous flown route matches; show one related-history reference; user opens past Trip/note; returns to current flight.	Dismiss hides the cue for this upcoming segment. Current alerts stay authoritative.
Optional loyalty setup	User opens Loyalty pilot; chooses program; optionally enters tier, balance and as-of date; explicitly chooses a personal review date.	The app labels entries User entered. Skip leaves core experience complete.
After travel	Operational flight completes; Trip becomes Eligible to close only after boundary checks; user closes or reviews; stats recompute; optional note.	Scheduled time passing alone never confirms flown status or credits loyalty balances.
Future pre-booking scenario	Outside MVP: user compares externally found offers with verified fare/loyalty inputs.	Present evidence and uncertainty; user books externally. Missing inputs mean no recommendation.

Trip recognition and user intent are separate. A flight provider can report a flight landed, but cannot prove that this account holder boarded. Proposed history uses the existing personal-flown signal when available, otherwise requests confirmation before including uncertain records in completed statistics.

13 Functional requirements: entry and grouping

Requirement	Trigger / input	Behavior and output	User action / errors
F01.1 Contextual entry	First Passport visit after feature eligibility; availability flag.	Show one dismissible invitation below primary history content. Record exposure only when visible.	Explore opens Trips; Not now suppresses invitation for 30 days. No modal on app launch.
F01.2 Data readiness	Trip view opens; local personal flight cache.	Render cache first. Explain history coverage and import access without implying all travel is present.	Retry failed sync. Existing history stays usable.
F02.1 Propose trips	Import or meaningful flight date/route/status change.	Run versioned deterministic rules over eligible unassigned flights. Persist candidates separately from saved Trips.	Review, edit or dismiss. Cancel candidate computation without changing flights.
F02.2 Review	Candidate selected; segment IDs and reason code.	Show title, dates, each leg and missing gaps. No silent saved Trip in MVP.	Save commits one Trip; deselection recomputes preview. One remaining leg is allowed.
F02.3 Manual create	User selects one or more personal flight records.	Chronological order is fixed by timestamps; preview gaps/overlaps. Preserve user-selected boundaries.	A time overlap requires explicit acknowledgement; invalid/missing identifiers block save.
F02.4 Dismissal	Not a Trip on a candidate.	Store rejection fingerprint using account + ordered flight IDs + rule version. Same unchanged set remains suppressed.	User can manually create or explicitly rerun suggestions; do not nag after every sync.
F02.5 Import resilience	Bulk import with duplicates or incomplete times.	Deduplicate exact known identities. Ambiguous identities remain separate and marked for review. Date-only records excluded from automatic grouping.	Failure yields an error summary and retained successful records; no all-or-nothing data deletion.

All core requirements are P0 unless labeled otherwise. Each must be traceable to the acceptance cases and feature flag list. Empty strings, unknown data and failed providers must never be presented as zeros or “no travel.”

14 Functional requirements: editing and retrieval

Requirement	Trigger / input	Behavior and output	Edges / failure behavior
F03.1 Rename	Edit title; 1-80 Unicode characters after trimming.	Save user title separately from generated title; do not overwrite on flight changes.	Blank invalid; display inline error. Long strings wrap in detail and truncate accessibly in list.
F03.2 Merge	Select 2-10 saved Trips.	Preview all unique legs in order; user chooses title and one cover. Retain source Trip IDs in operation history.	Combine notes with dated separators up to 2,000 characters; if over limit require manual revision, never truncate silently.
F03.3 Split	Select boundary between two legs.	Create two Trips; first retains title/note/photo by default; second gets generated title and no media. Preview allows relocating or copying note.	Two nonempty results required. Source flights unaffected.
F03.4 Move / remove leg	Select a leg in Edit.	Move to one destination Trip or ungroup; confirm affected Trip counts before atomic commit.	A segment belongs to at most one saved Trip. Empty Trip removal requires explicit confirmation.
F03.5 Ungroup / undo	Delete Trip container or reverse last edit.	Ungroup leaves flights intact; warn that Trip-specific note/image will be removed. Store reversible operation for 30 days.	Immediate Undo for 10 seconds; Recent changes recovery thereafter. Privacy-delete operations bypass recovery.
F04.1 Trip detail	Open saved Trip.	Display original records, gaps, next leg, completed/planned state and supported aggregates.	Canceled leg remains in timeline with canceled label but contributes no flown totals.
F04.2 Search	Search title, airport/city, airline or user note.	Case-insensitive literal/token search within authorized local data; AND across distinct filter categories.	No matches shows active chips and Clear filters; notes are never sent to general search services.
F04.3 Statistic drill-down	Tap supported airline, airport, route or year aggregate.	Open the exact contributing-flight IDs and current filter context. Trip grouping is a presentation option over those results.	Never include an entire Trip in the count when only one leg matches the filter.
F04.4 Source change	A source flight date/route is corrected or deleted.	Recalculate derived fields; mark saved Trip Needs review if route continuity changes. Keep manual boundary and note.	Deleted source becomes a recoverable missing reference until acknowledged; do not silently remove user context.

15 Functional requirements: reusable context and memory

Requirement	Trigger / input	Behavior and output	Edges / errors
F05.1 Related history	Upcoming personal flight within 14 days; prior confirmed flown record.	Match exact departure and arrival airport pair in the same direction within the past 24 months. Prefer latest match whose saved Trip has a note, otherwise latest match. Show one reference.	Do not match reverse route or same-city alternate airport in MVP. No match means no card.
F05.2 Explain relevance	Open reference.	Show “You flew this route in [month/year]”, original Trip and optional note with its timestamp. Link back to current flight.	Past gate, schedule or note is not presented as current advice. No reliability ranking from personal samples.
F05.3 Suppression	Dismiss / active disruption / pilot flags.	Suppress dismissed cue for this upcoming leg; hide it during active-flight disruption view and within 24h of departure.	Preparation value must not interrupt time-critical travel. No push notification for this cue in MVP.
F06.1 Note	Add or edit note on saved Trip.	Optional plain text, max 2,000 Unicode characters. Autosave local draft, explicit Save commits synced value.	On failure retain draft and show Retry; no AI rewrite or inferred purpose.
F06.2 Cover image	User selects one image through system picker.	Accept JPEG/PNG/HEIC up to 20 MB, normalize orientation, strip location metadata, create display copy capped at 2,048 px long edge.	No full-library scan. Permission denial leaves placeholder and all functionality intact. Upload failure does not block note/Trip save.
F06.3 Removal	Remove note/image.	Delete selected field only; confirm image removal where recovery is unavailable.	Source photo in the user’s photo library is never deleted. No media goes into analytics.
F07.1 Export	User selects Export my new travel data.	Export versioned JSON containing Trips, membership, source IDs, edits, notes and pilot snapshots; optionally package selected images in ZIP.	Exclude account tokens and third-party secrets. Confirm export destination in native share flow.
F07.2 Delete feature data	Explicit Delete Travel Profile data.	Remove new Trip/profile/pilot metadata and derived indexes; cancel pilot reminders; preserve original personal flights unless separately selected through existing deletion flow.	Clearly explain scope. Suppress automatic re-creation until user re-enables feature.

16 Trip recognition specification

Core distinction: a segment is one flight occurrence; a journey chain is a possible continuation; a Trip is a user-confirmed container. Automatic rules generate proposals, never facts about intent. “Confidence” means a rule category in v1, not a calibrated probability.

Configuration	Default	Reason / ownership
connection_gap_hours	8	Candidate continuity threshold aligned with documented connection convention; not a universal ticket-protection rule. ^[4]

Configuration	Default	Reason / ownership
stopover_gap_hours	24	Longer than 8h through 24h remains continuous but more ambiguous.
trip_gap_days	14	Maximum gap between non-connection legs for one automatic candidate. Proposed tunable heuristic.
trip_span_days	30	Maximum entire automatic candidate span. Longer trips remain manually groupable.
return_home_break_hours	24	A new departure after 24h at a confirmed home airport starts another candidate.
suggestion_visible_limit	3	Reduce proposal overload; More shows remaining candidates.
auto_save	False	Every proposed Trip requires confirmation in MVP.

Use scheduled departure/arrival in UTC for grouping, retaining IANA time zones and local dates for display. Real-time delays must not continuously regroup a journey. If scheduled timestamps are absent, date-only or inferred timezone records remain manual-review-only. Negative gaps and overlapping personal flights are ambiguous rather than automatically repaired.

- **1. Normalize:** filter to the account holder's own included flights. Exclude Friends, deleted and known not-flown records from completed candidates. Keep future scheduled flights only in upcoming candidates. Resolve only exact provider identity or verified codeshare equivalence.
- **2. Respect ownership:** remove all flights already assigned to saved Trips from candidate generation. Preserve all rejected boundary constraints. A changed source can flag a saved Trip for review but cannot dissolve it.
- **3. Build continuity:** same arrival/departure airport and a 0-8h scheduled gap is a strong continuation edge. A gap over 8h and at most 24h is a stopover edge. Different airports require a visible gap and review, even within one metro area.
- **4. Build a candidate:** extend within a maximum 14-day adjacent gap and 30-day span. A same-airport return with no unresolved gap is a likely roundtrip. With a confirmed home airport, close at the first return followed by at least 24h at home. Without home data, repeated returns to the starting airport are boundaries, not evidence of residence.
- **5. Resolve ambiguity:** multiple plausible partitions, home returns inside 24h, open-jaw gaps, relocation, missing legs or overlapping dates produce Needs review. Show at most one proposed partition; do not label a probability.
- **6. Name conservatively:** "[destination city] · [month year]" for a clear return; "Multi-city trip · [month year]" for ambiguity. Choose longest scheduled stay outside confirmed home; ties use earliest stop. Never infer work, romance, medical travel or other purpose.
- **7. Persist candidate:** record ordered IDs, reason codes, rule version, data revision and proposal hash. Save only after review. Re-running with unchanged inputs is idempotent.

Scenario	Expected grouping
PIT-ORD, ORD-HND; 7 days later HND-ORD, ORD-PIT	One likely roundtrip proposal, four segments; actual bookable examples only used as test fixtures.

Scenario	Expected grouping
PIT-HND; ground travel; KIX-SFO-PIT within 20 days	One Needs review proposal only if gap constraints hold. Dashed gap has no invented transport mode.
PIT-JFK-PIT in one day; PIT-ORD 2 days later	Same-day return is one candidate; later departure is another.
LHR-JFK returns locally earlier than departure clock time	UTC chronology remains positive; show both local dates/zones.
A traveler returns home after a trip, then leaves in 5 hours	Needs review; do not assume transit or a separate trip.
One-way relocation with a flight 2 months later	Separate proposals/manual record; never force a roundtrip.
Flight number reused for out-and-back or multi-stop service	Distinct occurrence IDs and airport pairs; flight number alone cannot deduplicate.
Canceled flight followed by replacement	Canceled timeline item may remain after user edit; only actual flown replacement contributes.
Unknown dates or a missing arrival time	Manual grouping only; suppress auto claims and duration total for missing fields.

Completing a Trip requires all included future legs to reach a terminal status and a confirmed end boundary. If there is no clear return, show End date not confirmed until the user closes it. “Landed” is a flight status; “Trip complete” is a separate container state.

17 Data model and calculation contract

Entity	Minimum fields	Invariant
FlightRecord reference	account_id, flight_id, provider_occurrence_id, ownership, inclusion_state, scheduled UTC times, IANA zones, airport IDs, operating/marketing carrier, status, flown confirmation, source revision	Provider occurrence, personal membership and marketing code are separate; never deduplicate by flight number alone.
Trip	trip_id, account_id, generated_title, user_title, state, start/end, boundary_confirmed, note, cover_asset_id, revision, created_at, updated_at	User title and boundaries override proposals; fields do not overwrite source flights.
TripMembership	trip_id, flight_id, ordinal, source_revision	Unique active (account_id, flight_id). Chronology derived; ordinal not arbitrary manual reordering.
TripCandidate	candidate_id, ordered_flight_ids, reason_codes, rule_version, source_revision, status, rejection_hash	Candidate cannot change an existing saved Trip.
ProfileAggregate	metric, filters, source_flight_ids, covered_count, missing_count, computation_version	Drill-down references exactly the contributing set.
EditOperation	operation_id, affected_ids, base_revisions, before/after, timestamp, expiry	Atomic commit with conflict detection; 30-day recoverability unless explicit privacy deletion.
LoyaltySnapshot (pilot)	program_id, tier_text?, balance_value?, balance_unit?, as_of_date, per_field_observed_at, source_type, user_review_date?, revision	Nullable means unknown; source_type=user_entered. No account number or password required.
LoyaltyRule (future)	program, qualifying_period, jurisdiction, effective dates, metrics, AND/OR paths, exclusions, official_source, verified_at, version	Expired/unverified rules cannot generate qualification advice.
ConnectedObservation (future)	provider, granted_scopes, observed_at, expires_at, field provenance, connection status	Never overwrite a manual observation without displaying conflict and freshness.

Completed profile set: unique included personal segments marked flown by the existing app/user, excluding cancellations, known no-shows, Friends and planned flights. Current-year filtering uses departure-local calendar year; a Trip spanning New Year appears in both matching search views but each flight is counted once in its own departure year.

- Airline distribution uses operating carrier; show marketing carrier separately. Unknown operating carrier is an explicit Unknown bucket, not silently reassigned.
- Distance uses the existing canonical distance field and labels its basis. If missing, show partial total with coverage; do not add planned or unrecorded ground distance.
- Flight time uses the existing canonical flown-duration field. Never mix airborne time and gate-to-gate time in one total. A new metric must use a distinct label.
- The exact query driving a total also drives drill-down. A 12-flight airline total opens 12 matching flights, even if their Trips contain 30 legs in total.
- Percentages use numerator / all eligible segments in that filter, including Unknown. Display n/N and coverage; do not describe a sample as all lifetime travel.

- Alliance statistics are V2: join operating carrier to an effective-date membership table for flight date. Alliance membership never proves crediting, status, upgrades or points eligibility.
- Route history uses exact direction and airports. Personal delay averages are not reliability predictions; suppress rankings in MVP.

18 Loyalty model, pilot and future decision support

The core product can succeed without loyalty setup. A **separate opt-in pilot** asks whether a basic user-maintained snapshot is valuable at all. Airline rules can distinguish marketed/operated travel, qualifying spend and excluded fares; Delta's published rules demonstrate why a completed flight count is insufficient to compute elite qualification.^[27]

Data category	Permitted interpretation	Forbidden inference
Existing flight history	Flights, routes, dates and documented operating airline.	Credited program, account balance or qualifying points.
Derived profile	Observed carrier distribution with denominator.	"You earned these miles" or "you are this tier."
User-entered snapshot	"You entered this tier/balance as of [date]."	Live synchronization, official verification or automatic balance increment.
Connected account (future)	Provider-supplied value, exact scope and observed time.	Universal coverage or truth beyond the provider's update time.
Third-party rules (future)	Versioned explanation with official source and applicability.	Eligibility without fare class, ticketing/crediting and relevant conditions.

L01 requirement	Behavior / error handling
Entry	User chooses Loyalty from profile settings only when pilot is enabled; no required onboarding step.
Inputs	Program from controlled catalog or Other; tier free text up to 60 characters; optional nonnegative integer balance; balance unit and as-of date required when balance is entered. Future as-of dates are invalid.
Output	Program + tier + user-updated date. Balance is secondary and labeled User entered. Unknown tier/balance displays Not added, never zero.
Freshness	After 30 days display "Not updated recently"; this is a UX reminder threshold, not a claim of airline validity. Editing one field updates only its observation date.
Reminder	A user may set one personal review date/time and explicitly enable a notification. It says "Review your [program] record," not "your miles expire." Reschedule replaces the previous notification.
Action	Open an allowlisted official program URL; if unavailable, show Copy program name and retain the snapshot. No automatic sign-in or sending account data in URL parameters.
After Trip	Do not increment any balance, segment requirement or tier. Suggest reviewing only if the user configured a reminder.
Exit	Delete snapshot cancels reminders and removes it from export/search; disabling pilot retains private data until the user deletes it.

Future qualification contract: redeemable miles, qualifying dollars/points, qualifying flight counts, tier validity and certificate expiration are separate types. Store qualifying-year boundaries and alternate AND/OR paths. Unknown required metrics produce Incomplete, not Near status. Do not reuse the brief's United example of 18/24 flights and 9,420/12,000 PQP as a current rule: it is unverified, and even on a hypothetical two-condition path both shortfalls matter.

Future offer example, not an MVP feature: "Option B costs \$50 more. Its qualifying contribution is unknown because fare class and crediting program are missing." Only verified inputs permit a scenario calculation. Show total cash cost, duration, stops and user preferences before loyalty. Never assign a universal dollar value to status, recommend unnecessary travel or claim a transferable-points balance proves award-seat availability.

19 State, sync and failure contracts

Object / transition	Rule	Visible recovery
Candidate: proposed to confirmed/dismissed	Confirmation creates Trip atomically; dismiss stores suppression.	Retry is idempotent by operation ID.
Trip: upcoming to in progress	At least one included personal leg starts; no automatic membership rewrite.	User can edit container; operational status remains sourced.
Trip: in progress to completed	All legs terminal, flown state known and end boundary confirmed.	Open-ended trips ask user to close; canceled-only trip is canceled, not flown.
Trip: any to needs review	Source deletion/change breaks chronology or continuity.	Highlight affected leg, retain title/note, offer Edit or Acknowledge.
Sync: local saved to syncing to synced	Commit local changes first with durable operation queue.	"Saved on this device" until confirmed. Offline remains usable.
Concurrent merge/split	Compare base revision on all affected Trips and memberships; all-or-nothing commit.	Keep local draft; show latest version and let user reapply. Do not last-write-wins membership.
Concurrent note edits	Preserve both revisions when overlapping fields change.	Choose a version or manually combine; no silent loss.
Provider unavailable	Keep cache and explicit last-updated time.	Retry on next foreground/network recovery; no repeated background error banners.
Partial media failure	Trip and text save independently of optional image.	Retry image or remove pending image; no duplicate uploads.
Rollback	Disable exposure/proposals with flags, retain saved user data.	Existing saved Trips remain readable/exportable. Never destructive schema rollback.

Client retries use exponential backoff with jitter, capped at 60 seconds while foregrounded; background behavior follows existing app scheduling. Duplicate mutations share an operation ID. A server validation error is not retried automatically. Flight updates and profile sync use distinct failure channels so a profile error cannot suppress flight alerts.

20 AI opportunities and limits

The MVP grouping engine is deterministic. AI is not required to prove the hypothesis and cannot substitute for data access. A later classifier may help rank ambiguous trip partitions only after a labeled evaluation set exists. All low-confidence partitions still require confirmation.

Opportunity	Appropriate use	Evaluation / limit
Trip clustering	Rank plausible candidate boundaries from authorized structured flight data.	Hold out travelers, not random flights; measure false joins, false splits and edit effort separately.
Anomaly detection	Flag impossible overlaps or likely duplicate imports.	Never delete records automatically; optimize review usefulness.
Natural-language explanation	Optional paraphrase of an already-validated structured reason.	Template fallback; generated text may not introduce dates, benefits or facts absent from input.
Loyalty forecast	Only a future validated rules engine can calculate eligibility.	An LLM cannot infer missing balances or invent qualifying rules.
Personal summaries	Optional, user-requested synthesis of selected notes in a later release.	Separate consent and clear source boundaries; no sensitive-purpose inference.

Before any model deployment, measure improvement against the deterministic baseline, latency, cost per processed account and unsupported-claim rate. Require zero invented financial/eligibility claims in the release test suite. If improvement is limited to wording, prefer templates. The current MVP sends no notes or photos to an external model.

21 Privacy, trust and non-functional requirements

The extension should follow Flighty's existing privacy posture rather than assuming a new cloud identity or unrestricted mailbox access. Its published policy describes specific server processing and analytics; it does not justify calling every new feature on-device or end-to-end encrypted.^[28]

Area	Proposed requirement / acceptance
Data minimization	Reuse existing flight records. Do not collect passport numbers, airline credentials, card numbers or full reward account numbers for this feature.
Consent	Explain new grouping on entry; require user selection for photos and explicit action for loyalty reminders. Future connections require separate scopes and revocation.
Isolation	Every read/write/export checks account ownership. Never aggregate Friends into personal history. Shared devices use existing account protection.
Transport/storage	Use existing authenticated encrypted transport and protected device storage. Security review must confirm server encryption and access controls; do not promise unverified E2EE.
Analytics	Use pseudonymous account/experiment IDs and categorical properties. No raw notes, images, names, account balances, precise routes or loyalty numbers in event payloads.
Retention	Proposed: edit history 30 days; event-level experiment data 90 days; feature metadata until user deletion. Purge live deleted content within 7 days and backups within 30 days, subject to confirmed infrastructure feasibility before launch.

Area	Proposed requirement / acceptance
Export	JSON schema version and user-owned metadata available without additional payment; image export explicit. Existing underlying provider-data export terms require review.
Performance	On an iPhone 13-class reference device with 2,000 cached segments: warm Trips list p95 under 1s; filter response p95 under 500ms; initial grouping under 5s off the UI thread. Proposed budgets, not measured results.
Operational guardrail	No changes to flight-alert scheduling; active-flight entry task time non-inferiority margin 10% versus baseline.
Availability	Cached Trips, notes and drill-down work offline. External program links and unsynced data show unavailable status.
Accessibility	Dynamic Type, VoiceOver labels, 44pt action targets, 4.5:1 body-text contrast and non-color status cues are explicit design acceptance targets.
Localization	Externalized strings and plural rules; IANA zones; locale-aware units and dates. Airport code always paired with readable city where space permits.

These are proposed engineering policies, not legal conclusions about a jurisdiction. If deletion or export timing cannot be honored with the existing architecture, resolve that gap before public release and write the actual policy into product copy.

22 Visual design and screen specifications

Evolve Flighty's aviation-focused hierarchy: route codes, compact flight rows, a globe/map layer and native sheets. Use restrained neutral surfaces with a single blue interaction color. Color communicates state, not decoration. Avoid scrapbook themes, heavy gradients, fintech dashboards, decorative gauges and gamified loyalty progress.

Element	Specification	Why
Typography	Use existing product text styles; large route/title, readable body, monospaced/tabular numeric treatment where appropriate.	Preserves hierarchy and scanning; no independent branding system.
Spacing	Proposed 8pt base rhythm; 16pt sheet side padding; comfortable flight-row touch area.	Dense data remains usable on mobile.
Trip route	Actual included flight legs as solid paths; unrecorded gap as dashed path with text.	Prevents invented continuity.
History reference	Small date-stamped reference below upcoming-flight operational content; no urgency color.	Distinguishes previous experience from current status.
Metadata labels	User entered, Based on 12 flights, Partial history, Saved on this device.	Makes evidence and sync state understandable.
Empty state	Specific explanation plus one action; never fake personal data.	Reduces setup confusion.
Editing	Preview structural changes before saving; clear Cancel and Undo.	Supports control, error prevention and recognition. ^[29]

Suggested screen content is shown below as text wireframes, not screenshots of the current app. Dates, counts and personal notes are illustrative test data.

Passport / Trips Your trips Suggested trip Japan · May 2026 5 flights · Review boundaries Review trip Saved trips Chicago · April 2026 2 flights · Personal note added Overview Trips	Review trip Japan · May 2026 These flights may belong together. PIT → ORD → HND Travel not recorded HND to KIX KIX → SFO → PIT Save Trip Edit flights · Not a Trip	Before you fly You have flown this route PIT → ORD · April 2026 Your note from April “Leave more time for breakfast.” Past trip details may have changed. Open previous trip Back to upcoming flight
--	--	--

Map labels and chronology should be validated against the actual test fixture before high-fidelity design. The illustrative five-leg open-jaw example is distinct from the earlier four-leg closed-roundtrip fixture.

Copy key	Default English	Use
trip.candidate.reason	These flights may belong together.	Proposal explanation; never asserts intent.
trip.gap.unknown	Travel not recorded	Missing leg or ground gap.
trip.save.local	Saved on this device	Offline commit awaiting sync.
history.reference.warning	Past trip details may have changed.	Old versus current context.
loyalty.snapshot.source	User entered · Updated {date}	Snapshot provenance; localize date separately.
trip.edit.conflict	This trip changed on another device. Review the latest version.	Conflict with retained draft.
profile.coverage	Based on {known} of {total} flights	Partial metric coverage; use plural rules.

23 Metrics and event specification

Primary outcome: repeat source-flight retrieval from history within 28 days. Count accounts opening a past personal flight detail from a history context on at least two distinct days. In control, that context is existing Passport/history; in treatment it also includes Trips, drill-down and related-history references. Instrument the same terminal flight-detail event in both arms before randomization. Trip saves, corrections, notes and Trip-only revisits are secondary meaningful-use diagnostics, not substitutes for the comparable primary endpoint. Seeing a card or opening Passport alone does not qualify.

Metric	Denominator / window	Proposed gate
Activation	Exposed eligible accounts saving a first Trip within 7 days / exposed eligible accounts.	At least 40% in the invited pilot; diagnose import/discovery friction if lower.

Metric	Denominator / window	Proposed gate
Repeat history use	Assigned eligible accounts retrieving a past source flight from history on at least two distinct days in 28 days / all assigned eligible accounts.	Directional target +5 percentage points versus control, with uncertainty reported.
Next-trip return	Assigned accounts returning to current-flight detail from 24h before to 6h after next scheduled flight / assigned accounts with an observed next flight within 90 days.	Secondary; no negative movement. Stratify by prior travel frequency and plan.
Grouping quality	Reviewed likely candidates accepted without membership edits / all reviewed likely candidates.	At least 85%; also measure reviewer refusal/abandonment and audit unreviewed examples.
Retrieval usability	Participants completing known-trip retrieval / assigned participants; time among successes separately.	At least 80% unassisted; median under 30s in the moderated study.
Preparation usefulness	Users rating related history useful or very useful / respondents who opened it.	At least 60% as a diagnostic; report response rate and actual return behavior.
Loyalty maintenance	Pilot snapshot creators updating or reviewing on a later day within 30 days / snapshot creators.	At least 40% to justify further research; not a core launch gate.
Non-travel use	Accounts making a meaningful action outside any known flight interval $\pm 24h$ / assigned eligible accounts.	Diagnostic only; missing imported travel can misclassify non-travel time.
Paid conversion	New paid subscriptions within 30 days / free accounts randomized and eligible at assignment.	Secondary exploratory endpoint; separate trials from paid, existing subscriptions and lifetime plans.
Churn / renewal	Cancellations and realized nonrenewals / renewable subscriptions reaching a renewal opportunity.	Report 90-day and renewal-aligned outcomes; annual churn cannot be proven in a 4-week test.
Guardrails	Core flight task time; crash-free users; sync/data loss; unwanted notifications.	No confirmed data loss; no meaningful alert regression; time margin under 10%.

Event	Required non-sensitive properties	Meaning
profile_entry_viewed	experiment, cohort, entry_surface, has_history	Visible exposure, not background loading.
trip_candidate_reviewed	candidate_class, segment_count_bucket, rules_version	Review denominator.
trip_candidate_resolved	action=accept/edit/dismiss, reason_category	Quality and rejection.
trip_saved / trip_edit_committed	creation_source, operation_type, leg_count_bucket	Server-confirmed or durable local action with event_id dedupe.
profile_drilldown_opened / source_flight_opened	metric_type, filter_count, coverage_bucket	Navigation then actual source use.
history_reference_opened / dismissed	match_type, prior_age_bucket	Preparation signal.
note_saved / image_attached	content_present, save_state	No content, photo metadata or note length beyond coarse bucket.

Event	Required non-sensitive properties	Meaning
loyalty_snapshot_saved / reviewed	pilot_group, field_presence, age_bucket	No raw program balance or tier.
export_completed / sync_failed	schema_version, error_category	Trust and recovery.

Use stable event IDs and UTC timestamps, with a coarse account-local day derived using the fixed experiment timezone chosen at assignment. Deduplicate offline retries. Freeze cohort eligibility at randomization. Analyze intention-to-treat for causality; activated-user analyses are descriptive and must not replace the primary comparison.

24 Research and experiment plan

Stage	Design	Decision
Discovery · week 1	12 interviews: 6 multi-airline users, 2 single-airline frequent users, 2 occasional users, 2 former Flighty users; include at least 4 international travelers across groups. Reconstruct the last real trip and an actual history lookup.	Proceed if at least 7 describe a recent retrieval/preparation task and at least 5 demonstrate material friction. Thresholds guide judgment, not population estimates.
Prototype · week 2	8 participants; counterbalance baseline and redesign; tasks: retrieve known trip, fix open-jaw split, explain airline total, find old note, return to active flight.	At least 80% core-task success, no repeated misinterpretation of old data as live, and no critical navigation blocker.
Grouping audit · weeks 2-3	200 consented or carefully constructed itineraries, separately label real vs synthetic; cover closed/open-jaw, time zones, duplicates, relocation, crew and cancellation. Hold out travelers.	At least 85% likely-proposal precision with false joins separately reported; no data-destructive behavior. Synthetic accuracy is not user acceptance.
Core A/B · after quality gates	Randomize account-level eligible users to current experience or core Trips+drill-down+history reference. Stratify by plan and travel frequency. Run at least 28-day follow-up.	Primary repeat-use lift, confidence interval and guardrails; do not declare success from activation alone.
Component follow-up	If core bundle is useful, compare core with/without related-history cue. Run loyalty as a separate opt-in randomized pilot versus no pilot invitation.	Separates preparation and loyalty value from trip organization.
Commercial test · later	After utility, test existing-Pro value explanation at a natural import/depth moment with actual billing events.	No new price or paywall claim until a powered conversion test.

Illustrative power planning only: if the baseline repeat-use rate were 20%, detecting 25% at two-sided 5% significance and 80% power needs roughly **1,100 accounts per arm** before attrition, using a standard two-proportion approximation. The 20% baseline is invented for planning, not observed. Recalculate with internal baseline, eligibility, clustering and missingness before running. A small class project cannot establish commercial lift; report usability and feasibility honestly.

Interview prompts: “Show me the last time you looked for a past flight.” “What did you do with that information?” “What did the airline app already solve?” “When did you last update a reward balance?” “What made you stop paying?” Avoid asking only whether a presented feature sounds useful.

Stop conditions: repeated data loss or active-flight interference blocks release. If retrieval improves but repeat use does not, ship a modest Passport usability improvement and drop the retention narrative. If users do not maintain manual loyalty snapshots, do not treat account integrations as the automatic solution; first establish a valuable task with a realistic data source.

25 Acceptance test matrix

Test	Given / when	Then
T01 Clear return	Four timestamped own flown legs form a return within limits.	One likely proposal; no saved Trip until confirm; all four source IDs retained.
T02 Open jaw	Arrival HND, next departure KIX.	Visible unrecorded gap; needs review; no invented transport.
T03 Time zones	International date-line crossing and DST boundary fixtures.	UTC order and elapsed time correct; local dates/zones readable.
T04 Duplicate import	Same canonical occurrence reimported; separate codeshare alias also present.	Verified identity counted once; unknown aliases flagged, not guessed away.
T05 Reused number	Same flight number on two routes/dates.	Distinct flight occurrences preserved.
T06 User correction	User splits proposal and sync reruns.	Saved memberships remain split; unchanged rejected grouping does not reappear.
T07 Merge undo	Merge two Trips with notes then undo.	Original IDs/memberships/content restored; aggregate flown totals unchanged.
T08 Over-limit note	Merge would exceed 2,000 characters.	Save blocks with editable preview; no truncation or lost note.
T09 Cross-year	Trip departs Dec 30 and returns Jan 3.	Trip discoverable in both periods; per-year flight totals use departure-local year.
T10 Filter lineage	Carrier filter matches 12 flights in larger Trips.	Total and source list both equal 12; no inflated Trip-wide count.
T11 Cancellation	Canceled flight plus flown replacement.	Canceled label retained if included, no flown contribution; replacement counted once.
T12 Missing data	Arrival timestamp or operating airline missing.	Partial duration or Unknown carrier shown; no false zero or qualification.
T13 Cross-account	Friend flight or another account ID enters request.	Authorization rejects it; no private data in output/events.
T14 Offline conflict	Two devices split/merge overlapping Trip memberships.	Atomic conflict surfaced; local draft preserved; no double membership.
T15 History match	Upcoming PIT-ORD, previous ORD-PIT only.	No reference. Same-direction eligible prior flight does produce reference.
T16 Active flight	Eligible history cue, but departure within 24h or active disruption view.	History cue hidden; flight task unaffected.
T17 Loyalty uncertainty	Snapshot missing balance; a Trip completes.	Not added remains; balance/tier never auto-increments.
T18 Permissions	Photo picker denied or upload fails.	Trip/note still save; accessible placeholder and Retry.
T19 Delete / export	User exports then deletes new feature data.	Readable versioned export; feature data/reminders removed per policy; source flights preserved.

Test	Given / when	Then
T20 Accessibility	Largest supported text size, VoiceOver and reduced motion.	No clipped primary action; reading order and non-map route equivalent usable.
T21 Entitlement	Pro lapses after a saved Trip.	Owned metadata remains readable/editable/exportable; existing provider/import limits are applied without silent deletion.
T22 Analytics	Offline action is retried three times.	One event after deduplication; no raw routes, notes or balances in payload.

QA sign-off requires representative physical-device testing and sync-failure fixtures. These are specified acceptance tests, not a claim that an implementation exists or has passed them.

26 Dependencies, roadmap and release gates

Illustrative implementation estimate: 6-8 weeks with one product designer/PM, two iOS engineers, part-time backend/data support and QA, assuming usable flight identity and sync infrastructure. This is an estimation basis, not a committed Flighty staffing plan. A course prototype can cover the core flow in two weeks using clearly labeled fixtures.

Phase	Scope / dependencies	Exit gate
0 · Evidence and feasibility · week 1	Current-build inventory; field access; sync ownership; baseline metrics; interviews. Owners: PM, research, engineering.	A01/A04 confirmed; no already-shipped duplicate proposition; supported retrieval task.
1 · Design and prototype · week 2	Trips, review/edit, drill-down and history reference; accessibility states. Owner: design.	Core usability target; old/live distinction understood.
2 · Data and core implementation · weeks 3-5	Identity, candidate rules, atomic edits, aggregation, offline sync and notes. Owners: iOS/backend.	Acceptance cases T01-T14; no data loss.
3 · Preparation and hardening · weeks 5-6	Related history, image handling, export, telemetry, entitlement and performance. Owners: iOS, QA, data.	Full acceptance suite and operational guardrails.
4 · Controlled pilot · weeks 7-8 onward	1% then 5% then 25% eligible rollout, with account holdout; allow sufficient observation windows.	Quality first, then 28-day outcome maturity. Eight-week build is not eight-week retention proof.
P1 · Separate loyalty pilot	Manual snapshot/reminder; independent flag and measurements.	Maintenance/usefulness evidence before any integration procurement.
V2 / future	Share recap, alliance taxonomy, one licensed reward connector, later rule-based scenarios.	Consent, coverage, economics and accuracy gates; never assume airline/bank access.

Dependency	Owner role	Fallback
Stable flight identities and own/flown flags	Engineering	Manual Trip creation on clearly confirmed records; defer automation.
Current feature/entitlement inventory	Product	Implement only verified gaps; preserve existing access.

Dependency	Owner role	Fallback
Atomic sync and deletion support	Backend/security	Block production rollout; prototype locally with fixtures.
Airport/timezone/reference data	Data engineering	Date-only manual handling and visible unknown labels.
Analytics randomization and billing outcomes	Data analyst	Report usability results only; no causal business claims.
Loyalty catalog and official link upkeep	Product operations	User-entered program name with no automatic rule engine.

Flags: `travel_profile_enabled`, `trip_suggestions_enabled`, `history_reference_enabled`, `loyalty_pilot_enabled`, `media_enabled`. A kill switch stops exposure and computation but retains user-owned records for reading/export. Stop rollout on confirmed data loss, cross-account leakage or operational alert regression. Do not wait for statistical significance to stop a critical failure.

27 Risks, open decisions and case-study narrative

Risk / open question	Mitigation / next decision
A better log may still not justify a subscription.	Keep conversion exploratory and test paid value after utility.
Manual loyalty setup may create work rather than reduce it.	Optional pilot; kill on weak maintenance and no clear task.
“Cross-airline” could sound unique when it is not.	Show Triplt/AwardWallet competition and narrow the claim to retrieval/reuse in Flighty.
History gaps create misleading certainty.	Coverage labels, partial totals, no inferred missing legs or earned points.
Same-airport rules fail for commuting, crew and relocation.	Conservative proposals plus manual control; exclude unsupported auto segments without blocking manual use.
Current build may already address the gap.	Release-day inventory is required; revise scope rather than defend the old concept.
Home airport may be sensitive or wrong.	Optional, explicitly chosen, multiple airports allowed; no inferred residence.
Pro packaging remains unsettled.	First test utility with no new paywall. Preserve user-owned data on lapse.
Market growth claim exceeds evidence.	No market-share promise. Measure eligible adoption, retention and eventually attributable acquisition.
Connected rewards might require commercial agreements.	No integration in MVP; evaluate one program/provider at a time only after demand evidence.

Portfolio narrative: Start with an excellent episodic flight utility and an initial idea to make trip memories. Show the challenge: emotional value alone does not demonstrate repeated or paid utility. Introduce evidence for retrieving grouped travel and drilling into history, alongside counterevidence that live tracking remains valuable. Compare the six strategies and explain why the selected redesign centers on actionable records rather than a rewards super app.

Show three design decisions with their consequences: user-confirmed grouping protects intent; source-linked statistics make summaries inspectable; a narrowly matched prior-trip note tests reuse before another journey. Place loyalty context in a separate experiment. Present the actual evidence level, prototype results when available and remaining risks. Do not present proposed targets as outcomes.

Case-study claim appropriate now: “I used public product research and qualitative user evidence to define a testable extension of Flighty’s travel history, then specified its mobile workflows, data boundaries and validation gates.” The eventual claim “increased retention” requires a real controlled result.

28 Sources and scope coverage

Numbered references link to original sources. Product descriptions are publisher claims; reviews are qualitative user reports. All feature requirements, scenarios, thresholds and test plans are original proposals, not sourced claims of implementation.

1. **Flighty.** [Product overview](#). Undated; accessed 2026-09-10. Operational tracking and prediction positioning.
2. **Flighty.** [How does Connection Assistant work?](#). Undated; accessed 2026-09-10. Existing cross-carrier connection guidance.
3. **Flighty.** [Can I track my flight stats and all-time flight history?](#). Undated; accessed 2026-09-10. Existing Passport baseline.
4. **Flighty.** [How does Flighty count layovers and connections?](#). Undated; accessed 2026-09-10. Existing stop-counting definitions; redesign heuristics are separate.
5. **Flighty.** [What is Flighty Pro, and why should I subscribe?](#). Undated; accessed 2026-09-10. Plan and feature baseline, not a new pricing proposal.
6. **Triplt.** [Triplt Pro](#). Undated; accessed 2026-09-10. Itinerary-adjacent tools and Point Tracker.
7. **AwardWallet.** [All your Rewards & Travel in One Place](#). Undated; accessed 2026-09-10. Reward aggregation incumbent.
8. **point.me.** [A better way to book with points](#). Undated; accessed 2026-09-10. Redemption positioning; no actual offer recommended.
9. **United Airlines / Apple.** [United Airlines App](#). Live listing; accessed 2026-09-10. Publisher description of app workflow.
10. **Delta / Google Play.** [Fly Delta](#). Listing updated 2026-09-03; accessed 2026-09-10. Android publisher description; no iOS parity assumption.
11. **American Airlines.** [American Airlines app](#). Undated; accessed 2026-09-10. First-party execution and loyalty features.
12. **ANA.** [ANA Mileage Club](#). Undated; accessed 2026-09-10. AMC ecosystem, not a comprehensive app audit.
13. **FlightAware.** [Live Flight Tracker](#). Live page; accessed 2026-09-10. Operational tracking category.
14. **Flightradar24.** [Subscription plans](#). Live page; accessed 2026-09-10. Aviation mapping/history tiers.
15. **Seats.aero.** [What is dynamic vs saver pricing?](#). Undated; accessed 2026-09-10. Award-pricing domain; exact product coverage unverified.
16. **Google.** [Find plane tickets on Google Flights](#). Undated; accessed 2026-09-10. Search tradeoffs and external booking handoff.
17. **KAYAK.** [Help and customer support](#). Undated; accessed 2026-09-10. Verified category coverage; limited Trips detail.
18. **Chase.** [Ultimate Rewards](#). Undated; accessed 2026-09-10. Issuer-controlled redemption and transfers.
19. **American Express.** [American Express Travel](#). Undated; accessed 2026-09-10. Travel and reward context; no card recommendation.
20. **Flighty.** [Flighty vs. App in the Air](#). Undated; accessed 2026-09-10. Closure date/migration, sourced to a competitor.
21. **Aimee1980 / Apple App Store.** [Flight attendant review on Flighty listing](#). 2024-01-03; accessed 2026-09-10. Historical trip grouping request; not a current deficiency audit.
22. **Reddit r/flighty.** [Recently most airlines have a good Live Activities feature....](#) Displayed 3mo; accessed 2026-09-10. Mixed operational/substitution evidence.

23. **Reddit r/flighty**. [Passport click through on stats](#). Displayed 13d/12d; exact date unresolved. Direct drill-down requests.
24. **Reddit r/flighty**. [Do all of you pay for this app?](#). Displayed 8mo; accessed 2026-09-10. Price, episodic payment and Triplt substitution.
25. **Reddit r/flighty**. [How do you guys track down missing flights?](#). Displayed 1y; accessed 2026-09-10. Surviving comments; original post is deleted.
26. **AwardWallet / Google Play reviewers**. [AwardWallet: Track Rewards](#). Reviews 2024-11-19 and 2024-06-13. Selected positive/negative account-management evidence.
27. **Delta Air Lines**. [How to Get Medallion Status](#). Live rules; accessed 2026-09-10. Illustrates eligibility complexity; no universal threshold copied.
28. **Flighty**. [Privacy Policy](#). Undated; accessed 2026-09-10. Existing privacy posture and processing descriptions.
29. **Jakob Nielsen / NN/g**. [10 Usability Heuristics for User Interface Design](#). 1994-04-24; reviewed 2024-01-30. Control, error prevention, recognition and status visibility.

Project inputs: the supplied Markdown research prompt (uploaded 10 September 2026); earlier “Flighty_Mobile_Redesign_PRD_v0.2.html,” dated September 2026, used for concept continuity and reported visual references. These are project materials, not independent market evidence. No earlier unsupported statistics were reused.

Requested area	Coverage
Industry and ecosystem	Existing baseline, competitive landscape, evidence boundaries.
User validation	12-record qualitative ledger, six segments, composite persona, research plan.
Six strategic positions	Weighted and dependency matrices; one recommended direction.
Layered feature model	Trips and profile in MVP; loyalty pilot; wallet/Next Move gated to future.
Journey and detailed PRD	Screen inventory, requirements, grouping, data, states, AI, privacy and QA.
Business and validation	Explicit denominators, experiment design, commercial limits and roadmap.
Visual and portfolio narrative	Mobile screen specifications, illustrative wireframes and evidence-appropriate narrative.